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Department of Marine Sciences,
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A. Education

- 2017 : Doctor of Philosophy, Biotechnology, National Chung Hsing University, Taiwan
2013 : Master of Science, Marine Biotechnology, National Dong Hwa University, Taiwan
2008 : Bachelor of Science, Fisheries Technology, University of Riau, Indonesia

B. Professional Experience

- 2021 – current : Assistant Professor, Dept. of Marine Sciences, Universitas Padjadjaran, Indonesia
2018 – 2022 : Postdoctoral Associate, Dept. of Plant Biology, Rutgers University, USA.
2017 – 2018 : Postdoctoral Associate, National Chung Hsing University, Taiwan.

C. Publications (2014-Current)

• Climate Change and Marine Pollution:

1. **Buntora Pasaribu***, Christopher Vincent Mishaël Dilens, Muhammad Wahyudin Lewaru, Diah Ayuningrum, Mufti Petala Patria, Donny Juliandri Prihadi, Noir P. Purba, Mochamad Untung Kurnia Agung, Ismail Maqbul, Sulistiowati Sulistiowati (2026). '*Shotgun Metagenomic Dataset of Seawater Bacterial Communities from Pari Islands, Indonesia*', Microbiology Resource Announcements (Accepted). (*Corresponding author)
2. **Buntora Pasaribu***, Iskandar, Rita Rostika, Sheila Zallesa, Yuli Andriani, Izza M. Apriliani, Muhammad R.A. Mahendra, Fanny Kristiadi, Irfan Zidni, Sarawinda Hutagalung, Alexander M.A. Khan (2025). 'Morphological Characteristics of Turion Formation and Development in *Spirodela polyrhiza*', *Egyptian Journal of Aquatic Biology and Fisheries*, 29(4), pp. 1677-1689. doi: 10.21608/ejabf.2025.443812. (*Corresponding author)
3. Batool, F., Kurniawan, T.A., **Pasaribu, B.**, Kusworo, T.D., Silviana, S., Kusuma, H.S., Khan, M.I., Haddout, S., Choo, W.O., Iqbal, M.A. (2025). Removal of chromium from electroplating wastewater using chitosan-coconut shell composite: kinetics, thermodynamics and artificial neural networks modeling. *Int. J. Environ. Sci. Technol.* <https://doi.org/10.1007/s13762-025-06591-2>.
4. Ajeng Puspita, Mega Syamsudin, Fadli Syamsudin, Subiyanto, **Buntora Pasaribu**. (2025). 'Assessing the Impact of the 2020 – 2021 La Niña Event on Longtail Tuna (*Thunnus tonggol*) Migration Patterns Using MaxEnt Modeling', *Egyptian Journal of Aquatic Biology and Fisheries*, 29(3), pp. 341-361. doi: 10.21608/ejabf.2025.426902
5. Batool, F., Kurniawan, T.A., **Pasaribu, B.**, Kusworo, T.D., Silviana, S., Kusuma, H.S., Khan, M.I.,

- Haddout, S., Choo, W.O., Iqbal, M.A. (2025). Removal of chromium from electroplating wastewater using chitosan-coconut shell composite: kinetics, thermodynamics and artificial neural networks modeling. *Int. J. Environ. Sci. Technol.* <https://doi.org/10.1007/s13762-025-06591-2>.
6. David Del Vito., Donny J. Prihadi., **Buntora Pasaribu**, Chandra J. Koenawan., Syawaludin A. Harahap. (2025). 'Analysis of the Relationship Between Reef Fish Abundance and Coral Reef Cover in Karimun District, Riau Archipelago (Case Study: Pandan, Seranjau, Resam, Sekatip, Kelawa Besar, Kelawa Kecil, and Sebaik Islands)', *Egyptian Journal of Aquatic Biology and Fisheries*, 29(2), pp. 2769-2788. doi: 10.21608/ejabf.2025.424874
 7. Donny J. Prihadi, Syawaludin A. Harahap, **Buntora Pasaribu**, Wahyuniar Pamungkas, Febri Y. Simbolon. (2025). Exploring Coral Reef Distribution and Health at Post 1, West Bali National Park, *Egyptian Journal of Aquatic Biology and Fisheries*, 29(3), pp. 1-9. doi: 10.21608/ejabf.2025.425696
 8. Ajeng Puspita, Mega Syamsudin, Fadli Syamsudin, Subiyanto, **Buntora Pasaribu**. (2025). 'Assessing the Impact of the 2020 – 2021 La Niña Event on Longtail Tuna (Thunnus tonggol) Migration Patterns Using MaxEnt Modeling', *Egyptian Journal of Aquatic Biology and Fisheries*, 29(3), pp. 341-361. doi: 10.21608/ejabf.2025.426902
 9. Oktriono, K., Kurniawan, T.A., Meidiana, C., Wong, W.K., Wou Onn, C., **Pasaribu, B.**, Casilla, J.C.C., Abdulkareem-Alsultan, G. and Kusuma, H.S., (2025). Reinforcing synergy between circular economy in tourism and decarbonization in waste sector using digitalization: Case study in Taipei and Sukunan (Indonesia) in promoting carbon neutrality. *Waste Management & Research*, p.0734242X251326268.
 10. Ayuningrum, D., Shukor, M.Y., Patria, M.P., **Pasaribu, B.** and Sulistiowati, S. (2025). Exploring plastic-degrading enzyme producers from Ascidian-associated bacteria in Karimunjawa Islands, Indonesia. *Biodiversitas Journal of Biological Diversity*, 26(3).
 11. Alisha D. Azzahara, **Buntora Pasaribu***, Eri Bactiar, Yudi Nurul Ihsan. (2025). Distribution of heavy metals in surface seawater and sediment of Cirebon coastal area, West java. *Egyptian Journal of Aquatic Biology and Fisheries*. 29, 487-207. (*Corresponding author)
 12. Tri Dewi K. Pribadi, **Buntora Pasaribu**, Kalysta Fellatami, Jiang Mingguo, Azimah Ismail, Choerunnisa Febriani, Ching Fui Fui, Yudi Nurul Ihsan. (2025). Isolation and identification of Cadmium from contaminated coastal sediment in the Northern Indramayu, Indonesia. *Pakistan Journal of Biological Sciences*, 28.
 13. Alexander M.A. Khan, Ellen Barrowclift, Yi Xu, Gihoon Hong, Noir P. Purba, **Buntora Pasaribu**, Lantun P. Dewanti, M. Rudyansyah Ismail, Anqiq T, Per Berggren. (2024). Satellite observed oceanographic drivers of Mobulidae fisheries catch in Southeast Indian Ocean. *Journal of Sea Research*, 201, 1002534.
 14. Purba, N.P., Faizal, I., Christie, D., Munawir, B. Pratama, Darryl Valino., Marine Martasuganda., Titin Herawati., **Buntora Pasaribu***. (2024). Mapping the sources and strands of marine debris via particle modelling and In-situ sampling approaches in archipelagic countries. *Sci Rep* **14**, 25804. (*Corresponding author- Nature Publishing Group-Top journal 5%-IF 5).
 15. Herawati T, Damayanti W, Yuda GA, Ihsan YN, **Pasaribu B**, Harsono G, Marlina M, Suseno H, Mustopa RA, Karya FSW, Febriani C and Purba NP. (2024) Assessment of microplastic characterization and distribution from surface water and the seabed in the Flores Sea,

Indonesia. *Front. Mar. Sci.* 11:1440587.

16. Kurniawan, T. A., **Pasaribu, B.**, Kusworo, T. D., Wibisono, Y., Goh, H. H., Zhang, D., Mohyuddin, A., Ali, I., Aziz, F., Anouzla, A., Sarangi, P. K., Mahmoud, M., & Khan, M. M. H. (2024). Building disaster resilience in Thousands Islands (Indonesia): Unlocking climate adaptation strategies to navigate sea level rise in coastal regions while safeguarding crop productivity and local biodiversity. *ACS ES&T Water*, **4**, 3213–3224.
17. Tonni Agustiono Kurniawan, Christia Meidiana, Hui Hwang Goh, Dongdong Zhang, Mohd Hafiz Dzarfan Othman, Faissal Aziz, Abdelkader Anouzla, Prakash K Sarangi, **Buntora Pasaribu**, Imran Ali. (2024) Unlocking synergies between waste management and climate change mitigation to accelerate decarbonization through circular-economy digitalization in Indonesia. *Sustainable Production and Consumption*.46: 522-542 (Top journal 1%-IF 12)
18. Noir Purba, **Buntora Pasaribu**, Ibnu Faizal, Marine Martasuganda, Muhammad Ilmi, Choerunnisa Febriani, Raffy Alfarez. (2024) Coastal Clean-up in Southeast Asia: Lesson Learned, Challenges, and Future Strategies. *Frontiers in Marine Sciences* 10:1250736.
19. Syawaludin Harahap, Donny Prihadi, Ibnu Faizal, **Buntora Pasaribu**, Gusnadi Asri, Jiang Pei Luen. (2024) Day-night comparison of the number *Drupella* sp seen in The Semak Daun Island Coral Reef Seribu Islands, Indonesia. *AACL Bioflux* 17:50-60.
20. Yuniarti MS, Rizky Aulia Ramadhan, **Buntora Pasaribu**, Sheila Zallesa. (2024) Study of Surface Sediment Distribution on Grain Size in the Coastal Estuary of Marina Semarang, Central Java. *AACL Bioflux*.
21. **Buntora Pasaribu***, Noir Purba. (2023). Impact on Nuclear Waste of Fukushima Nuclear Power Plant to Indonesian Seas. *The conversation*. (Prestigious Scientific Newspaper in Indonesia) (*Corresponding author).
22. Lucy Omeyer, Emily M. Duncan, Neil Angelo S. Abreo, Jo Marie V. Acebes, Lea A. AngSinco-Jimenez, Sabiqah T. Anuar, Lemuel V. Aragonese; Gonzalo Araujo, Luis R. Carrasco; Marcus A.H. Chua, Muhammad R. Cordova, Lantun P. Dewanti, Emilyn Q. Espiritu, Jovanie B. Garay, Elitza S. Germanov, Jade Getliff; Eva Horcajo-Berna, Yusof S. Ibrahim, Zeehan Jaafar, Jose Isagani B. Janairo, Thanda Ko Gyi, Danielle Krebs, Cheng Ling Lim, Youna Lyons, Putu L.K. Mustika, Mei Lin Neo, Sirius Z.H. Ng, **Buntora Pasaribu**, Agamuthu Pariatamby, Cindy Peter, Lindsay Porter, Noir P. Purba, Ernesto T. Santa Cruz, Shahriar Shams, Kirsten F. Thompson, Daniel S. Torres, Rodney Westerlaken, Tuempong Wongtawan, Brendan J. Godley. (2023) Interactions between marine megafauna and plastic pollution in Southeast Asia. *Science of The Total Environment*. 874: 162502. (Top journal 5%-IF 10)
23. Yudi Ihsan, Kalysta Fellatami, **Buntora Pasaribu**, Tri Dewi Pribadi. (2023) Nutrient variability, pollution, and trophic status in Muara Gembong, Bekasi. *AACL Bioflux*. 16:1669-1681.
24. **Buntora Pasaribu**, I-Ping Lin, Chii-Shiang Chen, Chi-Yu Lu, and Pei-Luen Jiang (2014) Nutrient Limitation Induced Qualitative Changes of Fatty Acid and Caleosin Expression in *Auxenochlorella protothecoides*. *Biotechnology Letter*. 36:175–180.
25. **Pasaribu B**, Li YS, Kuo PC, Lin IP, Tew KS, Tzen JTC, Liao YK, Chen CS, and Jiang PL. (2016) The effect of temperature and nitrogen deprivation on cell morphology and physiology of *Symbiodinium*. *Oceanologia*. 58: 272-278.

- **Marine Policies:**

26. Prihadi Donny. J, Guanghai Zhang, Lahbar Ghulam M, **Buntora Pasaribu**. (2024) Integration of Community based Tourism (CBT) index and biophysical assessment for sustainable eco-tourism mangrove: A case study of Karangsong, Indonesia. Sustainability.
27. Alexander Khan, Ming-guo Jiang, Xiao-qiang Yang, Izza Mahdiana Aprilliani, Noir Purba, Budy Wiryawan, Am Azbas Taurusman, **Buntora Pasaribu***. (2024). Illegal Fishing threatens the sustainability of future Tuna commodities in Indonesia. Marine Policy. 159:105936. (*Corresponding author) (Top journal 3%-IF 3.8)

- **Cell and Molecular Biology in Plants:**

28. Muhammad, Z., Abdullah, **Pasaribu, B.**, Farooq, S., Khan, S., Ramzan, R., Mustfa, W. and Lamptey, F.P., (2026). Plant-Derived Oleosomes: Biotechnological Advancements on the Composition, Membrane Proteins, and Food and Pharmaceutical Applications. *Food Frontiers*, 7(2), p.e70244.
29. Ernst, E., Abramson, B., Acosta, K., Hoang, P.T., Mateo-Elizalde, C., Schubert, V., **Pasaribu, B.**, Albert, P.S., Hartwick, N., Colt, K. and Aylward, A., 2025. Duckweed genomes and epigenomes underlie triploid hybridization and clonal reproduction. *Current Biology*. 35, 1-20. (Cell Press-Top journal 1%-IF 8.1)
30. **Buntora Pasaribu***, Noir Primadona Purba, Lantun Paradhita Dewanti, Daniel Pasaribu, Alexander Muhammad Akbar Khan, Syawaludin Alisyahbana Harahap, Mega Laksmi Syamsuddin, Yudi Nurul Ihsan, Sofyan Husein Siregar, Ibnu Faizal, Titin Herawati, Mohammad Irfan, Timbul Partogi Haposan Simorangkir, Tonni Agustiono Kurniawan. (2024) Lipid Droplets in Endosymbiotic Symbiodiniaceae spp. Associated with Corals. *Plants*. 13: 949. (*Corresponding author). (Top journal 15%-IF 4.5)
31. **Buntora Pasaribu**, Kenneth Acosta, Bradley W. Abramson, Kelly Colt, Norlan, T. Hartwick, Yuanxue Liang, John Shanklin, Todd P. Michael, Eric Lam. (2023) Genomics of Turion from Greater Duckweed reveals its pathways for dormancy and reemergence strategy. *New Phytologist*. 239:116-131. (Top journal 5%-IF 10)
32. **Buntora Pasaribu***, and Pei-Luen Jiang. (2021) Agrobacterium-mediated Transformation of *Symbiodinium* spp. *Turkish Journal of Fisheries and Aquatic Sciences*. 21:291-298 (*Corresponding author).
33. Kenneth Acosta, Klaus J. Appenroth, Ljudmilla Borisjuk, Marvin Edelman, Uwe Heinig, Marcel Jansen, Tokitaka Oyama, **Buntora Pasaribu**, Ingo Schubert, Shawn Sorrels, Sowjanya Sree, Shuqing Xu, Todd P. Michael and Eric Lam. (2021) Return of the Lemnaceae: duckweed as a model plant system in the genomics and post-genomics era. *The Plant Cell*. 0:1-28. (Top journal 1%-IF 12).

- **Biochemistry and Biotechnology in Plants and Aquatic Organisms:**

34. Herawati, Titin, Indriyani Rahayu, Aisyah Aisyah, Mochamad Untung Kurnia Agung, **Buntora Pasaribu**, Atikah Nurhayati, Aadiana Ghazali, Roffi Grandiosa, Thallita Nasywa Faddilah, and Rendika Kamiswara. "Report on Vibrio Species Contamination in Shrimp From the Coast of

- Pangandaran, West Java, Indonesia." *Environmental Microbiology Reports* 17, no. 5 (2025): e70210.
35. Sofyan Husein Siregar, Efriyeldi , Zulkifli , Ronal Kurniawan, Ersya Fitriani, Agung Dhamar Syakti, Rihad Hakam, **Buntora Pasaribu**. (2025). Dataset on the Diversity of Epipellic Diatoms in Sediment Layers as Bioindicators of Aquatic Environment. *Egyptian Journal of Aquatic Biology and Fisheries*. 29 (2): 1187-1204
 36. Titin Herawati, Rahayu Indriyani, Aisyah Aisyah, Mochamad Untung K. Agung, **Buntora Pasaribu**, Atikah Nurhayati, Adiana Ghazali, Roffi Grandiosa, Thalita N. Faddilah, Randika Kamiswara. (2025). Contamination of pathogenic bacteria in the Vannamei Shrimp *Litopenaeus vannamei* pond area along the coast of Pangandaran, Indonesia. *Egyptian Journal of Aquatic Biology and Fisheries* (accepted)
 37. Rita Rostika, Fittrie Meylianawaty Pratiwy, Aulia Andhikawati, **Buntora Pasaribu**. (2023) The Addition of Fish Oil to Commercial Feed as Omega 3 Fatty Acid Source in Biofloc *Clarias* sp. Culture System. *AACL Bioflux* 16:55-65
 38. **Buntora Pasaribu**^{*}, Mike Chen, and Eric Lam. (2021) Optimizing a Protocol for Long-term Storage of Duckweed Clones as Turions: Organization of *Spirodela polyrhiza* in RDSC. Duckweed forum. (Non-peer reviewed) (^{*}Corresponding author).
 39. **Buntora Pasaribu**^{*}, Jou-Hsuan Fu, and Pei-Luen Jiang. (2020) Identification and Characterization of Caleosin in *Cycas Revoluta* Pollen. *Plant Signaling Behavior*. 8:1779486. (^{*}Corresponding author).
 40. **Buntora Pasaribu**, Miki Wang, and Pei-Luen Jiang. (2017) Identification of Oleosin-like Protein in Seagrass. *Biotechnology letters*. 9: 1757-1763.
 41. **Buntora Pasaribu**, Chii-Shiarng Chen, Yue Ken Liao, Pei-Luen Jiang, and Jason T.C. Tzen (2017) Identification of Caleosin and Oleosin in Oil Bodies of Pine Pollen. *Plant Physiology and Biochemistry*. 111:20-29. (Top journal 5%-IF 6.5)
 42. **Buntora Pasaribu**, Tse-yu Chung, Chii-Shiarng Chen, Pei-Luen Jiang, and Jason T.C. Tzen. (2016). Identification of Steroleosin in Oil Bodies of Pine Megagametophytes. *Plant Physiology and Biochemistry*. 101: 173-181. (Top journal 5%-IF 6.5)
 43. **Buntora Pasaribu**, Li-Chi Weng, I-Ping Lin, Eddie Camargo, Jason T.C. Tzen, Ching-Hsiu Tsai, Shin-Lon Ho, Mong-Rong Lin, Li-Hsueh Wang, Chii-Shiarng Chen, and Pei-Luen Jiang (2015). Morphological variability and Distinct Protein Profiles of Cultured and Endosymbiotic Symbiodinium cells Isolated from *Aipthasia pulchella*. *Scientific Reports*. 5: 15353. (Nature Publishing Group-Top journal 5%-IF 5)
 44. Li-Chi Weng[§], **Buntora Pasaribu**[§], I-Ping Lin, Ching-Hsiu Tsai, Chii-Shiarng Chen, and Pei-Luen Jiang (2014). Nitrogen Deprivation Induces Lipid Droplet Accumulation and Alters Fatty Acid Metabolism in Symbiotic Dinoflagellates Isolated from *Exaipthasia pulchella*. *Scientific Reports*. 4: 5777. ([§] These authors contributed equally to this work). (Nature Publishing Group-Top journal 5%-IF 5)
 45. **Buntora Pasaribu**, I-Ping Lin, Jason T. C. Tzen, Guang-Yuh Jauh, Tung-Yung Fan, Yu-Min Ju, Jing-O Cheng, Chii-Shiarng Chen, and Pei-Luen Jiang (2014) SLDP: a Novel Protein Related to Caleosin Is Associated with the Endosymbiotic Symbiodinium Lipid Droplets from *Euphyllia*

glabrescens. Marine Biotechnology. DOI 10.1007/s10126-014-9574-z.

46. Jiang PL, **Pasaribu B**, and Chen CS (2014) Nitrogen- Deprivation Elevates Lipid Levels in *Symbiodinium* spp. by Lipid Droplet Accumulation: Morphological and Compositional Analyses. PLOS ONE. DOI:10.1371/journal.pone.0087416.
47. **Buntora Pasaribu**, Tse-yu Chung, Chii-Shiarng Chen, Song-liang Wang, Pei-Luen Jiang, and Jason T.C. Tzen (2014). Identification of Caleosin and Two Oleosin Isoforms in Oil Bodies of Pine Megagametophytes. Plant Physiology and Biochemistry. 82: 142-150. (Top journal 5%-IF 6.5)

- **International and National (Books and Book Chapters)**

1. **Buntora Pasaribu***, Mohammad Kurnia Agung, Samuel Finley, Desniwaty Karo-Karo, Syahrul Aninullah, Haswan B.M. Harahap., Sahat M. Panggabean. (2024). Potential Application of Molecular Approach to Identify Invasive Non-Native Species (NNS) in Marine Litter. Lakisha Publisher. (*Corresponding author)
2. **Buntora Pasaribu***, Noir P Purba, Alex M. A. Khan. (2024). Genetic Transformation of Fruit and Vegetable Crops: Challenges and Opportunities. AAP Publisher. 398pp: ISBN-9781774917480. (*Corresponding author)
3. Microplastic in the Coastal Aquaculture Input Chain: From the Perspectives of Policy, Regulation, and Research to a Recommendation of a Mitigation Plan. Asia Pacific Economic Cooperation (APEC). (2024), 62pp. (as one of the authors in the book)
4. Yudi N. Ihsan, Tri Dewi K. Pribadi, **Buntora Pasaribu**, Lantun P. Dewanti, Noir P. Purba, Sanny T. Utami, Kalysta Fellatami, Agung T. Agustiana. (2023). Transboundary Diagnostic Analysis-Indonesia Seas large Marine Ecosystem ISLME. Food and Agriculture Organisation of the United Nation (FAO) Publisher. 337pp.
5. Yudi N. Ihsan, Tri Dewi K. Pribadi, **Buntora Pasaribu**, Lantun P. Dewanti, Noir P. Purba, Sanny T. Utami, Kalysta Fellatami, Agung T. Agustiana, Choerunissa. (2023). Strategic Action Plan Indonesia Seas large Marine Ecosystem ISLME. Food and Agriculture Organisation of the United Nation (FAO) Publisher. 120pp.

D. Presentation at International Meetings

Invited speaker/Oral presentations

1. The 10th Energy Oceania, Sydney, Australia, 2025
2. Mooring Training, IOCAS, China, 2025
3. The 8th China and ASEAN Marine Forum, Jakarta, Indonesia, 2024.
4. Marine Biogeochemistry Indonesia, Bandung, Indonesia, 2024.
5. The 5th International Conference on Science and Science Education, Salatiga, Indonesia, 2023.
6. Indonesian Journalist Workshop on Marine Plastic, Australian Embassy- Jakarta, 2023
7. The 2nd International Symposium on New Frontiers in Reef Coral Biotech, Taiwan, 2023
8. Professor Discovery to Protect Our Ocean-International Webinar, Bandung, Indonesia 2022
9. UKSW Department of Chemistry Webinar Series, Salatiga, Indonesia, 2022
10. International Conference for Fisheries and Marine Sustainability, Bandung, Indonesia, 2022.

11. Plant and Animal Genomics XXVIII, San Diego, California, USA, 2020.
12. Early Career Scientist Conference, Busan, South Korea, 2017.
13. Southeast Asia Workshop in Aquatic Microbial Ecology, Nha Trang, Vietnam, 2013.

C. Awards and Scholarship

2025	: Best Researcher in Indonesia and Indonesia Diaspora
2024	: Favorite Lecture in Perikanan Laut Tropis PSDKU Pangandaran
2023	: Winner of 115 Best Indonesia innovator 2023 in Renewable Energy
2015 - 2017	: Outstanding Publication Award - National Chung Hsing University, Taiwan
2015	: Outstanding Poster Award - Taiwan Coral Reef Society Meeting, Taiwan
2013 - 2017	: National Chung Hsing University Scholarship, Taiwan
2012	: Outstanding Academic Performance - National Dong Hwa University, Taiwan
2011 - 2013	: National Dong Hwa University Scholarship, Taiwan

G. Academic societies

1. Editorial Board Member, Scientific reports (Nature Publishing Group)- 2025 to now
2. Guest editor for special issue-Plants Journal (Basel) (2025)
3. Guest editor for special issue-Plants Journal (Basel) (2023)
4. Associate editor in Indonesian Scholar Journal (2013-2014)
5. Member of Taiwan Evolutionary Genomic and Computational Biology (2013)
6. Member of International Phycological Society (2011-2012)

H. Professional Services

- Manuscripts reviewer in top international journals including Current Plant Biology (Scopus Q1), Turkish Journal of Fisheries and Aquatic Sciences (Scopus Q2), Frontiers in Plant Sciences (Scopus Q2), Frontiers in Energy Research ((Scopus Q1), Science and Technology Indonesia Journal, Net Journal of Agriculture Science (Scopus Q3), and Aquatic Indonesia Journal ((Scopus Q4).
- Adhoc Scientific Technical Expert Members for Development Coastal and Marine Conservation Area Biawak Island, Marine and Fisheries Office of West Java-Republic of Indonesia. (2023)
- Adhoc Scientific Technical Expert Members for Development of Coastal and Marine Conservation Area Pangandaran, Marine and Fisheries Office of West Java-Republic of Indonesia. (2023)
- Technical training Program on Monitoring Methodologies of Marine Litter, Ministry of Coordinator and Investation Indonesia and Ministry of Environment Japan. (2023)
- Expert for Workshop APEC on Developing a Mitigation Plan to Prevent/Reduce Microplastics in Coastal Aquaculture Input Chains, Bali Indonesia, APEC. (2023)
- Expert Members for Project Development the Strategic Action Plan (SAP) for the Indonesia Seas Large Marine Ecosystem (ISLME), Project of Food and Agriculture Organisation (FAO) United Nations, supported by Global Environment Facility (GEF). (2023)
- Expert Members for Finalization of the Important Transboundary Diagnostic Analysis (TDA) for the

Indonesia Seas Large Marine Ecosystem (ISLME), Project of Food and Agriculture Organization (FAO) United Nations, supported by Global Environment Facility (GEF). (2023)

- Founder of TORGAS Laboratory, Marine Biogas Production and Conservation for Coastal Community (2022-now)

I. Intellectual Property Rights (IPR)

- Portable TORGAS for biogas production (Paten: *submitted*)
- Artificial Intelligence in Shrimp Aquaculture for Food Sustainability, Indonesia Copyrights No. 000858331, 10 February 2025
- Methods of measurements water quality for Lobster through Manual and Artificial Intelligence in Shrimp Aquaculture, Indonesia Copyrights No. 000860632, 13 February 2025.
- Identification of Oil Bodies in Lily Pollen, Indonesia Copyrights, August 2025
- Methods in Identification of Oil Bodies in Seagrass, Indonesia Copyrights, August 2025
- Turion Development, Indonesia Copyrights, August 2025

J. International and National Certification

- Open Water Diver Certification-Professional Association of Diving Instructor (PADI)
- Marine Conservation Instructor (Indonesia Expert Certification)
- Indonesia Water Pollution Assesment (Indonesia Expert Certification)
- Advanced Scuba Diver (International ADS)

K. Academic Supervision

I have supervised the thesis research of 25 undergraduate students and 3 graduate students, providing consistent academic guidance, methodological support, and mentoring throughout their research process.

L. Teaching Courses

Undergraduate courses:

1. Molecular biology (3 credits)
2. General Chemistry (3 credits)
3. Natural product (3 credits)
4. Organic Chemistry (3credits)
5. Marine Pollution (3 credits)
6. Microbiology (3 credits)
7. Coralogy (3 credits)
8. Fisheries Technology Process (3 credits)
9. Fisheries Packaging and Storage Technology (3 credits)

Graduate courses:

1. Advanced in Genetics and Molecular Ecology (3 credits)
2. Advanced Topics in Marine Biodiversity (3 credits)